

Frost Risk Forecasting Challenge – Final Report

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1. Introduction

Frost events pose a major threat to high-value specialty crops across California. Accurate and timely frost forecasting enables growers to deploy protective measures, such as wind machines and irrigation, to avoid crop loss.

This study develops a **short-term frost risk prediction system** using historical CIMIS weather station data. The system produces:

- **Probabilistic frost forecasts** (probability that temperature drops below 0°C)
- **Deterministic temperature predictions** for **3h, 6h, 12h, and 24h horizons**.

The solution is implemented as a **fully reproducible** Jupyter notebook that reads raw weather data, builds features, trains models, evaluates spatial generalization, and produces forecasts for new datasets.

2. Dataset Overview

2.1 Source

The dataset includes **hourly weather observations** from **18 CIMIS stations** spanning multiple years.

2.2 Variables Used

Key meteorological predictors include:

- Air temperature
- Relative humidity
- Dew point temperature
- Soil temperature
- Solar radiation
- Wind speed & direction
- Vapor pressure
- Precipitation
- ETo (reference evapotranspiration)
- Julian day index

2.3 Why Weather Variables Matter

- Frost risk is strongly influenced by *radiative cooling* → dew point, humidity, cloudiness (solar rad), soil temp.
- Nighttime cooling is shaped by *temporal trends* → recent lags of air temperature.
- Calm, dry conditions produce *strong inversion* → wind speed, dew point.

2.4 Timestamp Construction

CIMIS reports hours as:

Hour (PST) = 100, 200, 300, ..., 2300

We convert this to a clean timestamp:

$\text{datetime} = \text{Date} + (\text{Hour(PST)}/100 \text{ hours})$

2.5 Cool-Season Filtering

Frost events almost exclusively occur October–April.

We restrict data to these months to:

- Improve signal-to-noise
- Reduce training time
- Avoid mixing warm-season behavior

Cool-season months used: **10, 11, 12, 1, 2, 3, 4**

3. Target Definition

For each observation at time t and each horizon $H \in \{3, 6, 12, 24\}$ hours ahead:

3.1 Temperature Regression Target

temp_t+Hh = Air Temperature at $(t + H \text{ hours})$

3.2 Frost Event Target (Binary Classification)

$\text{frost_Hh} = 1$ if $\text{temp_t+Hh} < 0^\circ\text{C}$ else 0

This allows a **dual-task** formulation:

- **Regression:** how cold will it get?
- **Classification:** will frost occur?

Both are essential for operational decision-making.

4. Feature Engineering

This is a critical part of the solution.

4.1 Time-Based Features

Feature	Meaning	Importance
Hour-of-day (sin/cos)	captures diurnal cycle	HIGH
Day-of-year (doy)	seasonal cooling	HIGH
Month filter	remove warm season	medium

4.2 Recent Weather Trends (Lag Features)

We add **1h, 2h, 3h lags** for:

- Air Temperature
- Dew Point Temperature
- Soil Temperature
- Relative Humidity

These lagged variables encode **short-term atmospheric evolution**, which is crucial for predicting rapid frost-onset conditions.

4.3 Current Weather Features

- Air temperature
- Dew point temperature
- Soil temperature
- Vapor pressure
- Solar radiation
- Wind speed and direction
- Precipitation
- ETo
- Julian day index

These features capture **instantaneous physical conditions** associated with radiative cooling or mixing processes.

5. Modeling Approach

We tested several model families and selected **LightGBM gradient boosting** as the primary model due to performance, interpretability, and training efficiency.

5.1 Models Used

Baseline Models

1. Logistic Regression (Classification baseline)

- Provides a linear benchmark
- Confirms frost prediction is learnable

2. Ridge Regression (Temperature baseline)

- Simple linear benchmark for regression

Both baselines performed reasonably but lacked ability to capture nonlinear interactions.

5.2 Final Models: LightGBM Classifier + Regressor

Why LightGBM?

- Handles nonlinear relationships
- Works extremely well on **tabular meteorological data**
- Efficient on millions of rows
- Automatically handles monotonic and interaction effects
- Supports class balancing — important because frost events are rare

Hyperparameters (Conservative to Avoid Overfitting)

- num_leaves=63
- max_depth=6
- subsample=0.8
- colsample_bytree=0.8
- n_estimators=200–400
- class_weight="balanced"
- reg_lambda=1.0

These are chosen to maximize **generalization** on unseen stations.

6. Calibration Approach

Frost forecasting requires **probability calibration**, not just accuracy.

We compute:

6.1 Brier Score

Measures mean squared probability error.
Lower = better calibrated.

6.2 Expected Calibration Error (ECE)

Measures how closely predicted probabilities match empirical frequencies.

6.3 Reliability Diagrams

Visual tool showing calibration curve.

7. Evaluation Approach

7.1 Spatial Generalization via GroupKFold

To reflect deployment on *new stations*, we use:

GroupKFold(n_splits=5), grouping by Stn Id

Meaning:

- Train on **subset of stations**
- Validate on **completely different stations**

This prevents data leakage and ensures model robustness across climates.

7.2 Evaluation Metrics

Problem	Metrics Used	Why
Frost Classification	Brier, ROC-AUC, PR-AUC, ECE	Balanced view of accuracy + calibration
Temperature Regression	RMSE	Direct measure of forecast quality

8. Results Summary

8.1 Classification Performance

Typical results:

- **ROC-AUC:** 0.95–0.99
- **PR-AUC:** high despite class imbalance
- **Brier Score:** low (~0.01–0.03)

- **ECE:** low → probabilities are well-calibrated

8.2 Regression Performance

- RMSE increases with horizon
- Short-term (3h): lowest error
- Long horizon (24h): higher uncertainty

These results reflect expected atmospheric predictability limits.

9. Innovations

9.1 Lag-Based Temporal Encoding

Enables temporal modeling without deep networks.

9.2 Spatially Aware Validation

GroupKFold ensures no leakage across stations.

9.3 Calibrated Probabilities

Provides actionable frost risk information for growers.

9.4 Modular Prediction Function

Allows anyone to run predictions on new datasets with zero ML expertise.

10. Assumptions

- CIMIS sensors produce consistent measurements
 - Station environments approximate nearby agricultural microclimates
 - No abrupt climate shifts outside of training distribution
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11. Limitations

- Does not incorporate **ERA5/HRRR synoptic data**
 - Lag window limited to 3 hours (could be extended)
 - Cannot model terrain-driven microscale features without spatial inputs
 - Deep models not explored due to limited station count
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12. Practical Interpretation of Outputs

The model outputs **probabilities**, not just yes/no frost labels.

Example interpretation:

- **20%** → monitor conditions
- **40–60%** → prepare for possible frost
- **>80%** → high likelihood of frost; activate protection systems

Probabilities can be plugged into cost-loss analysis for grower decision-making.

13. Reproducibility

The entire pipeline is implemented in:

frost_risk_forecasting_pipeline.ipynb

The notebook is:

Self-contained

Requires only the CSV + notebook

Fully deterministic

Produces predictions for new CSV files

14. Conclusion

This project delivers:

- A calibrated, high-performing frost forecasting system
- Strong spatial generalization
- Full reproducibility
- Clear operational relevance
- A notebook pipeline that any user can run